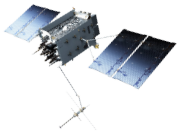


Localizing satellites with TDOA

Ignacio Ready
CU Boulder RF & SatNav Laboratory

Intro

GPS/GNSS jamming



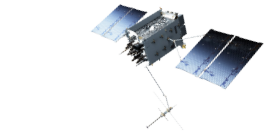
The US's GPS satellites are ~20,200 km above earth [1].

Their signals cover a huge region and are weak.

An emitter on Earth, broadcasting on the same frequency with high power, can drown out GNSS signals for receivers listening nearby.

Intro

Space-based jamming?



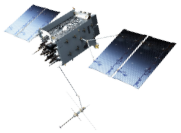
June 2, 2026: Researchers at UT Austin (Clements et al.) pre-publish a paper:

Chasing Lightning: Detecting, Characterizing, and Identifying a Powerful Space-Based GNSS Interference Source [2]

- Finds a satellite interfering with GPS over “Europe, Greenland, and Canada since 2019” [2]. Might be quick jammer tests from Russian satellites.

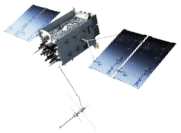
Because of distance, malicious satellite signals have much **wider geographic reach** than ground jammers.

Intro



Objective:

Analyze the feasibility of various methods to convert my lab's existing ground-based TDOA system into a space-jammer localization system.

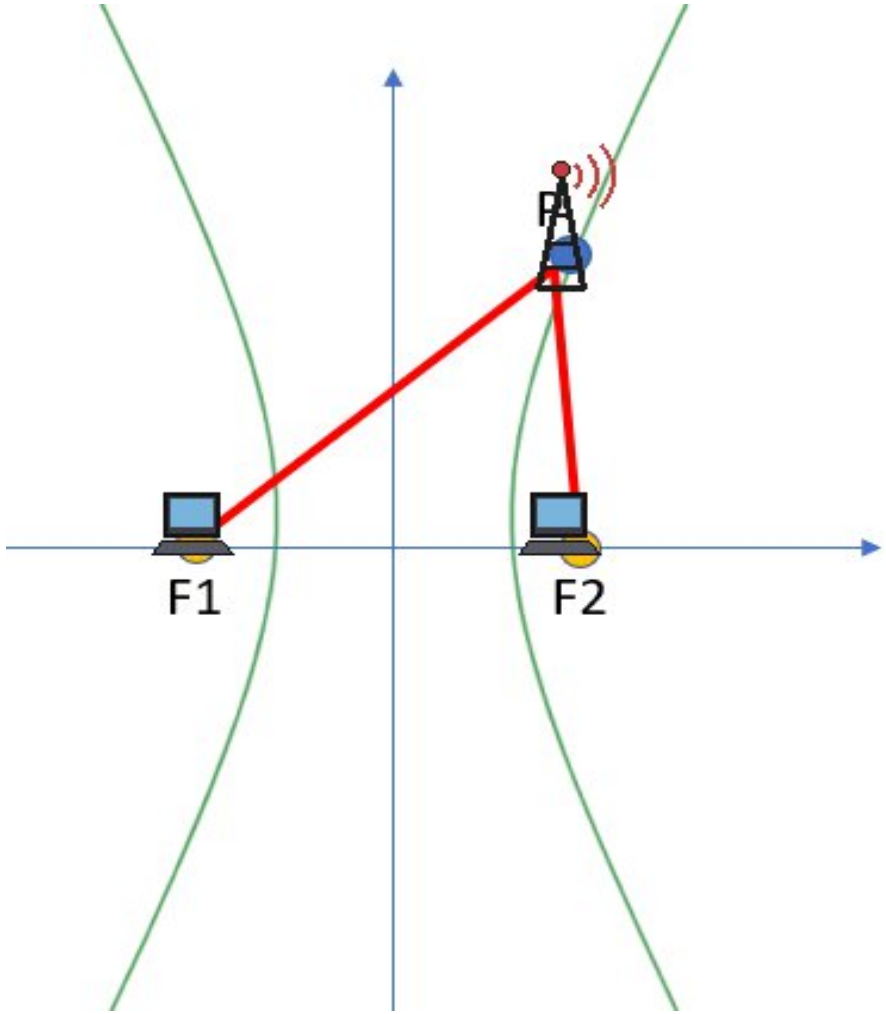


Existing research

How does the RF & SatNav lab localize sources of ground-based GPS jamming?

Existing work

TDOA = Time Difference Of Arrival



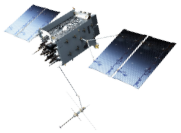
- If a radio signal arrives at F2 200 nanoseconds before F1:

The emitter must lie on **hyperbola** such that the **distance difference** between F1 to P and F2 to P must equal 200ns

- Many nodes ➞ many TDOA pairs ➞ many hyperbolas ➞ constrained solution

Existing work

NLLS: Solving **realistic** 2D TDOA

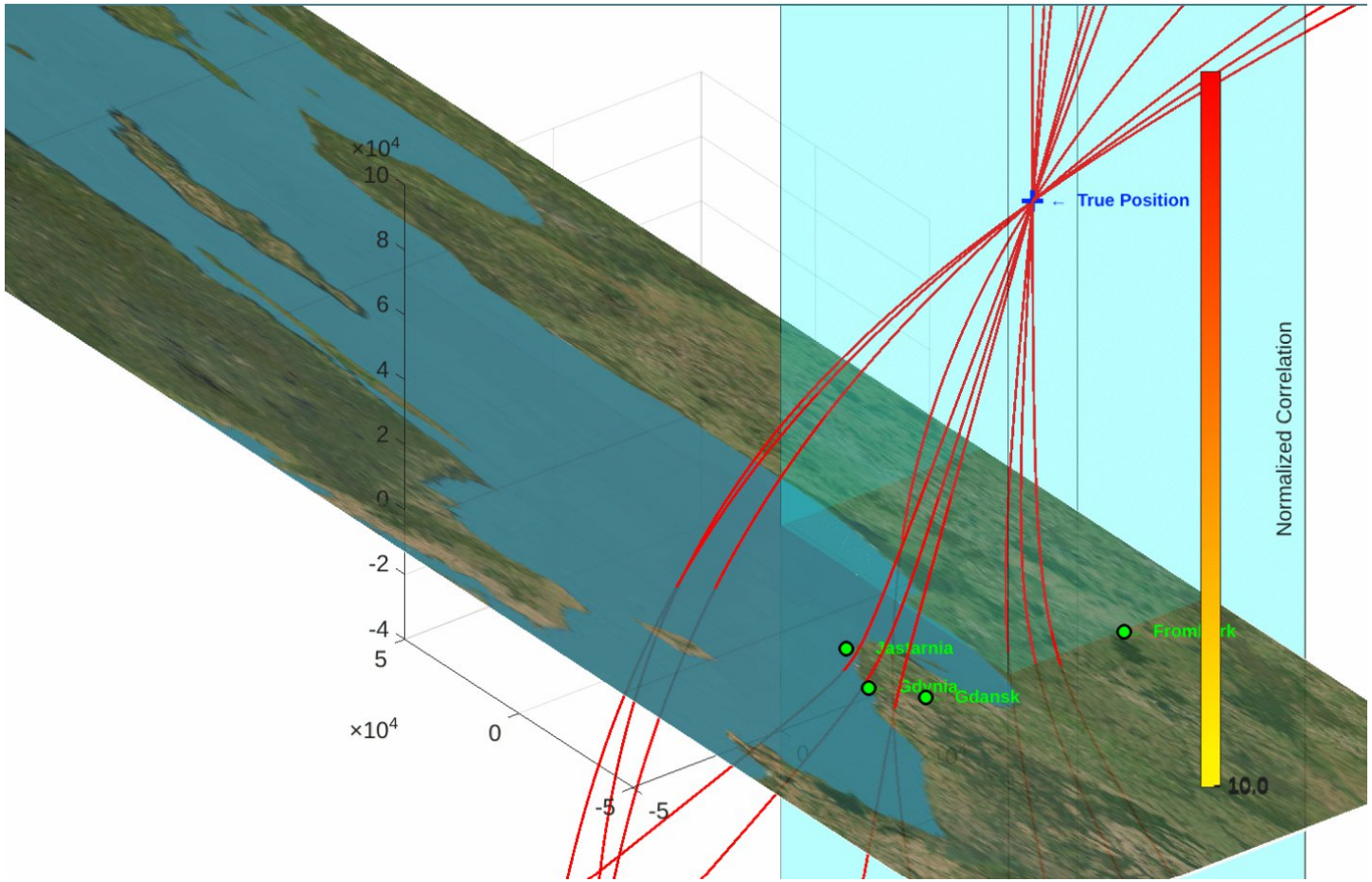


Non**L**inear **L**east **S**quares

- The hyperbolas ***don't*** all intersect at one point; what 2D point satisfies them all the ***best***?
- Guess a point inside box
- Measure 2d distance from guess point to each line (squares ➞ Euclidean distance; least ➞ closest)
- Keep guessing points and shrinking box until you minimize distance(residuals).

Attempt #1: NLLS

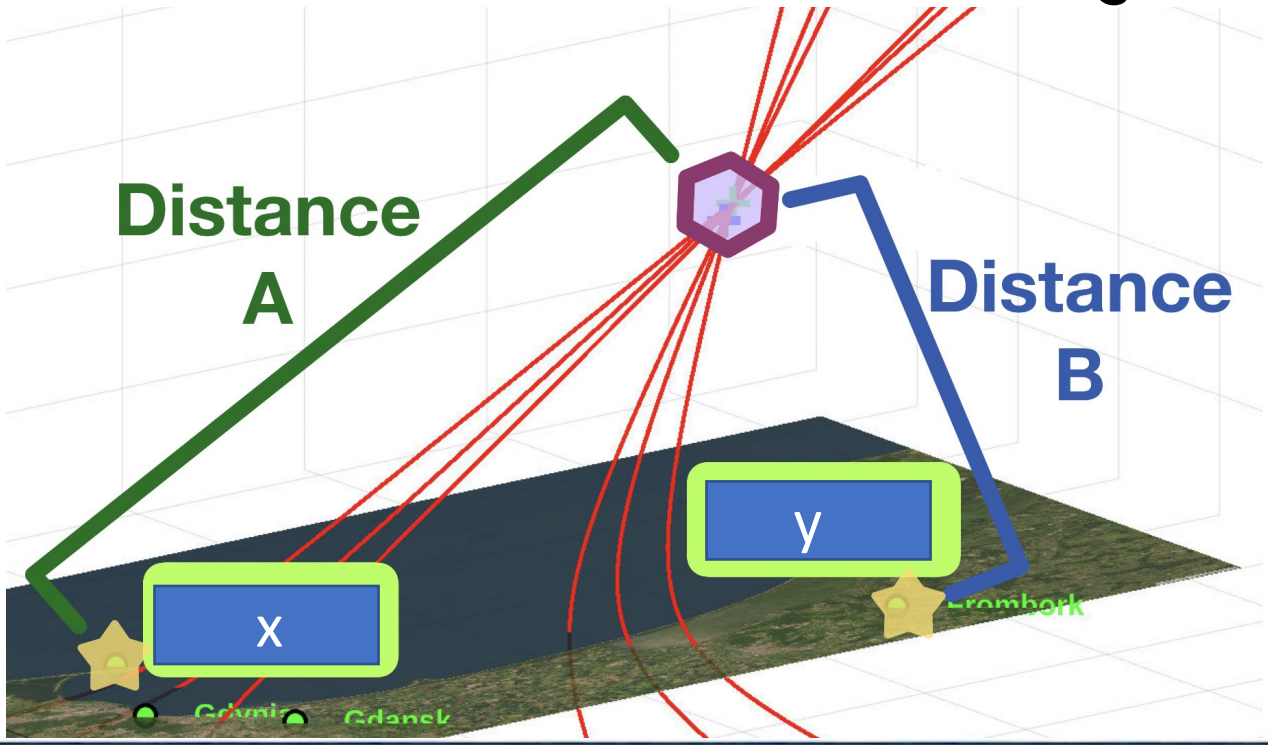
Adapt lab's existing ground-based NLLS and TDOA algorithms to solve for the 3D position of a jammer satellite.



Attempt #1

Method: Part 1

- Choose artificial jammer location
- Calculate 3D distance from each node jammer
- Calculate **time-difference** of signal arrival for each **node pair** (TDOA)





For x-y node pair:

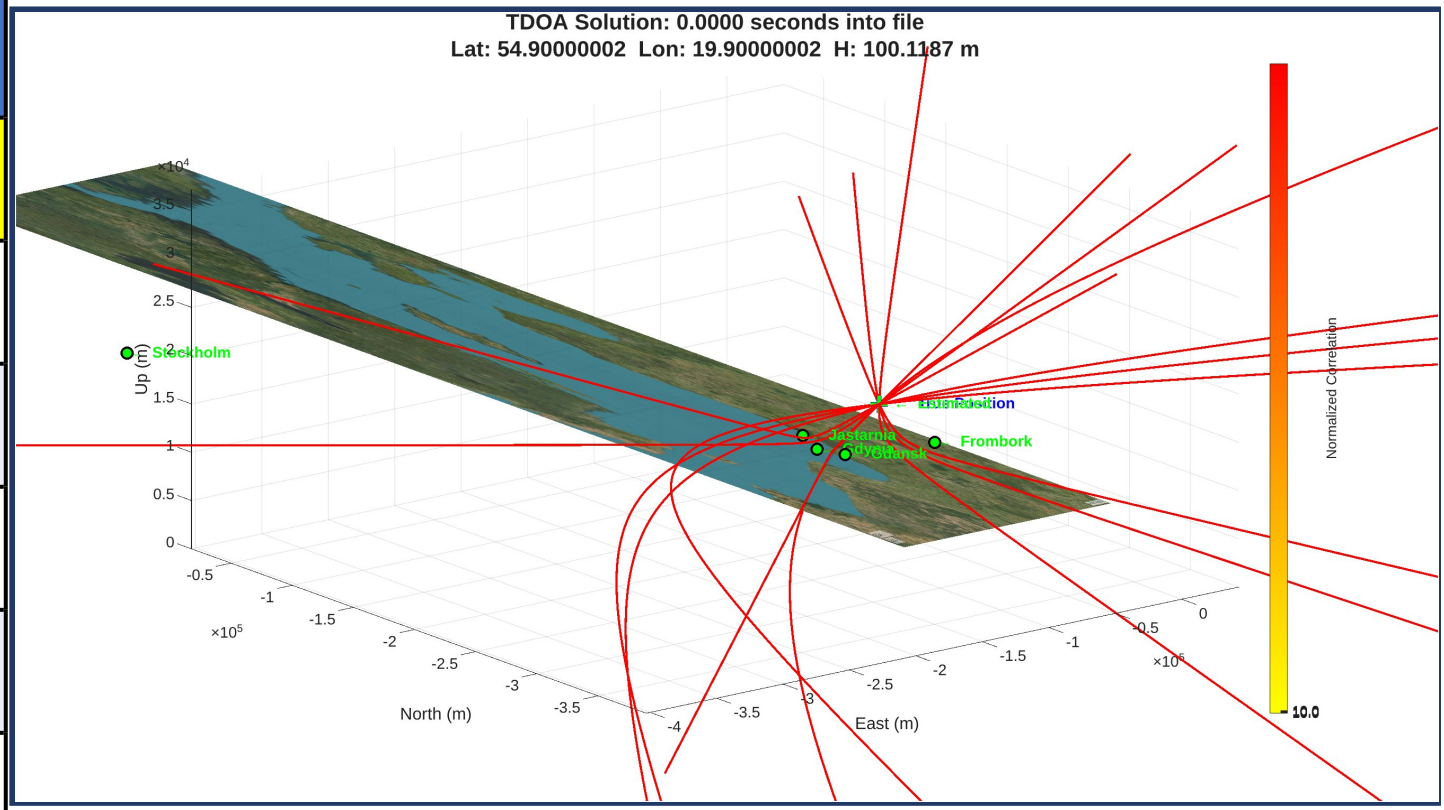
ΔDistance ÷ (speed of light) = TDOA

Attempt #1

Results

NLLS with perfect TDOA

Real height (km)	Calc. Height (km)	Height Error (m)	3D Error (m)
0.1	0.1	0.12	0.12
10	-7.79	-17792.54	17795.02
100	100	0.03	0.04
1,000	1,000	0.044	0.05
10,000	10,000	-0.09	0.09
30,000	30,000	0.4	0.4
50,000	50,000	-0.36	0.36

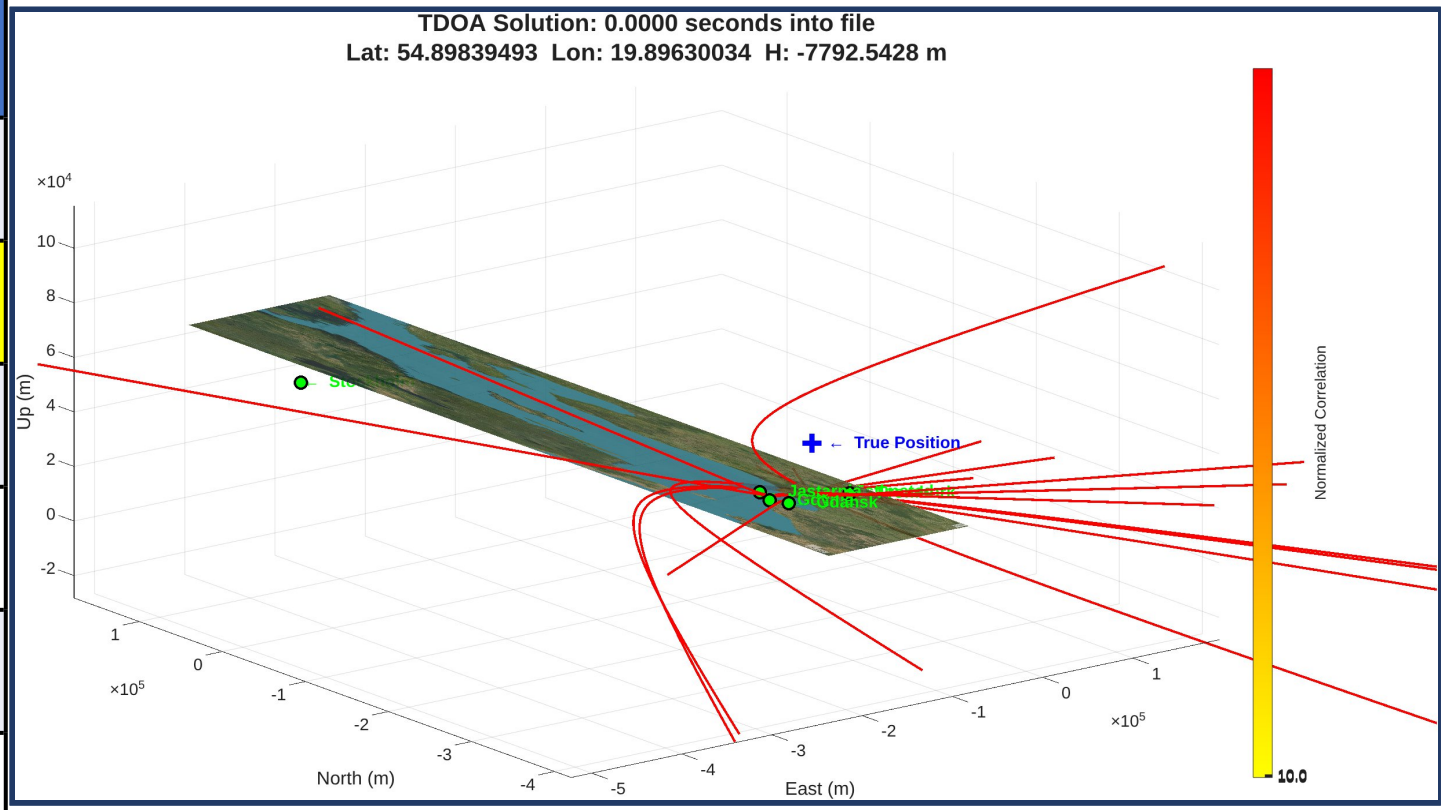


Attempt #1

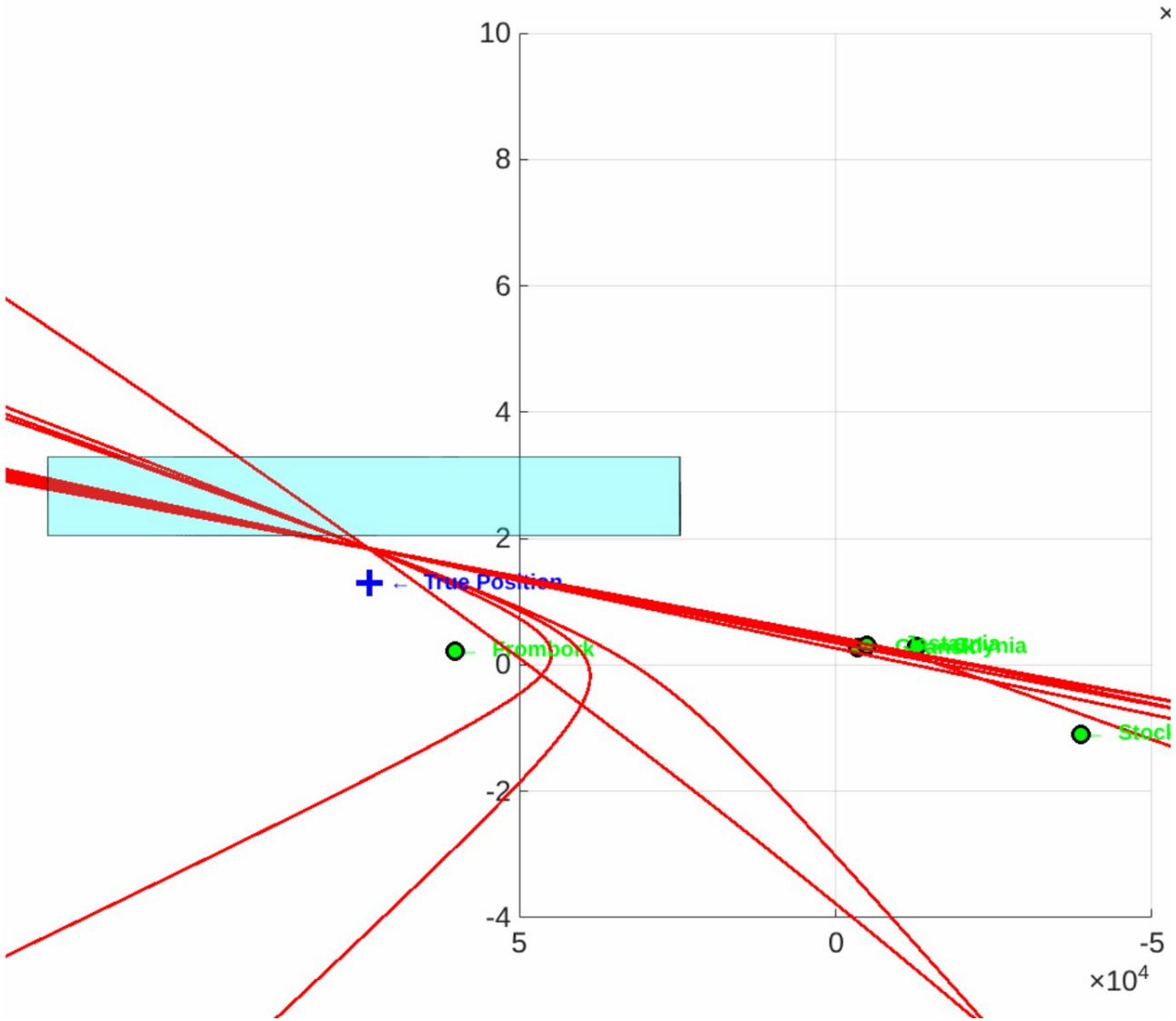
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50,000	50,000	-0.36	0.36



NLLS messes up if parameters are poorly tuned.

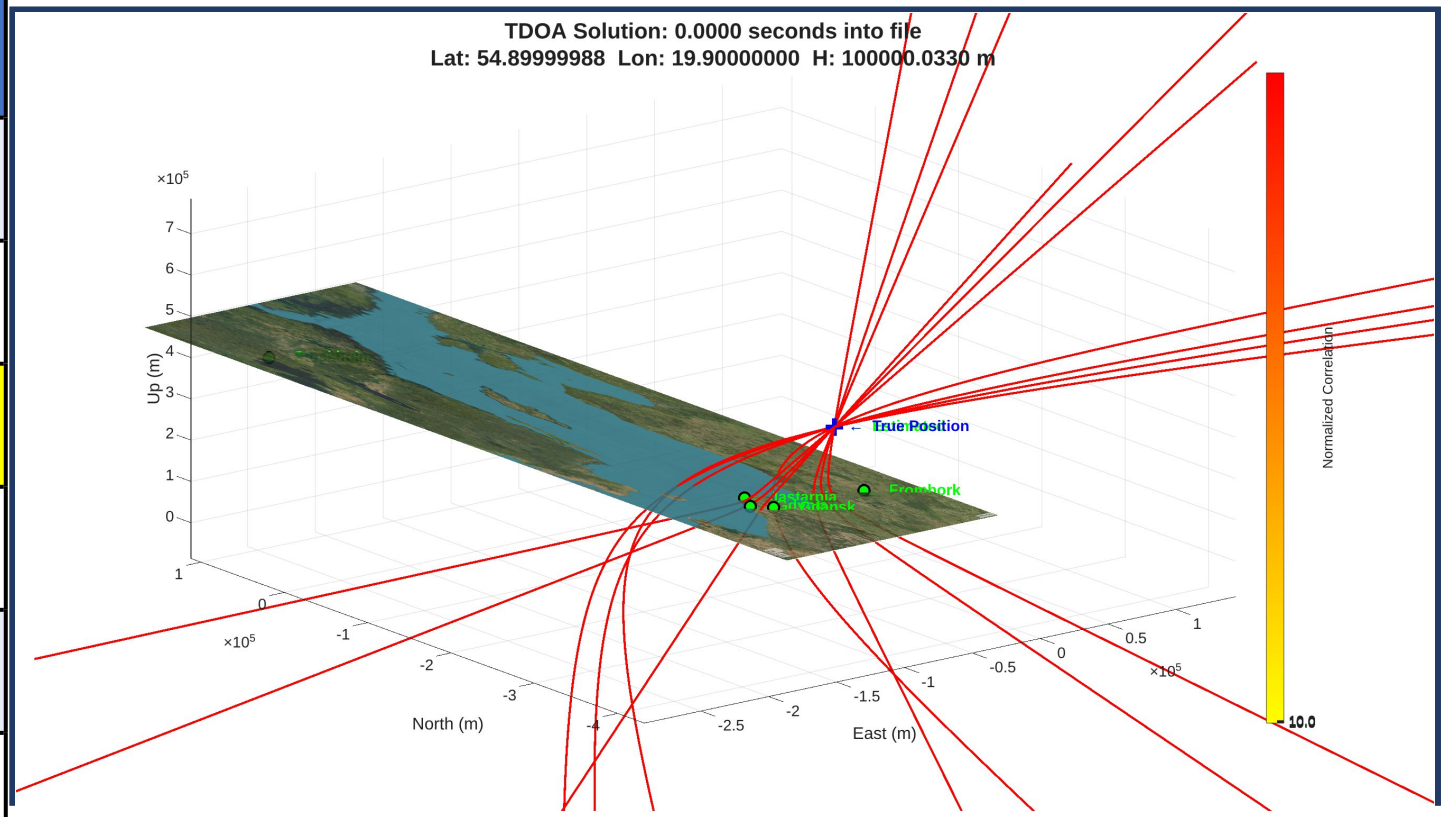


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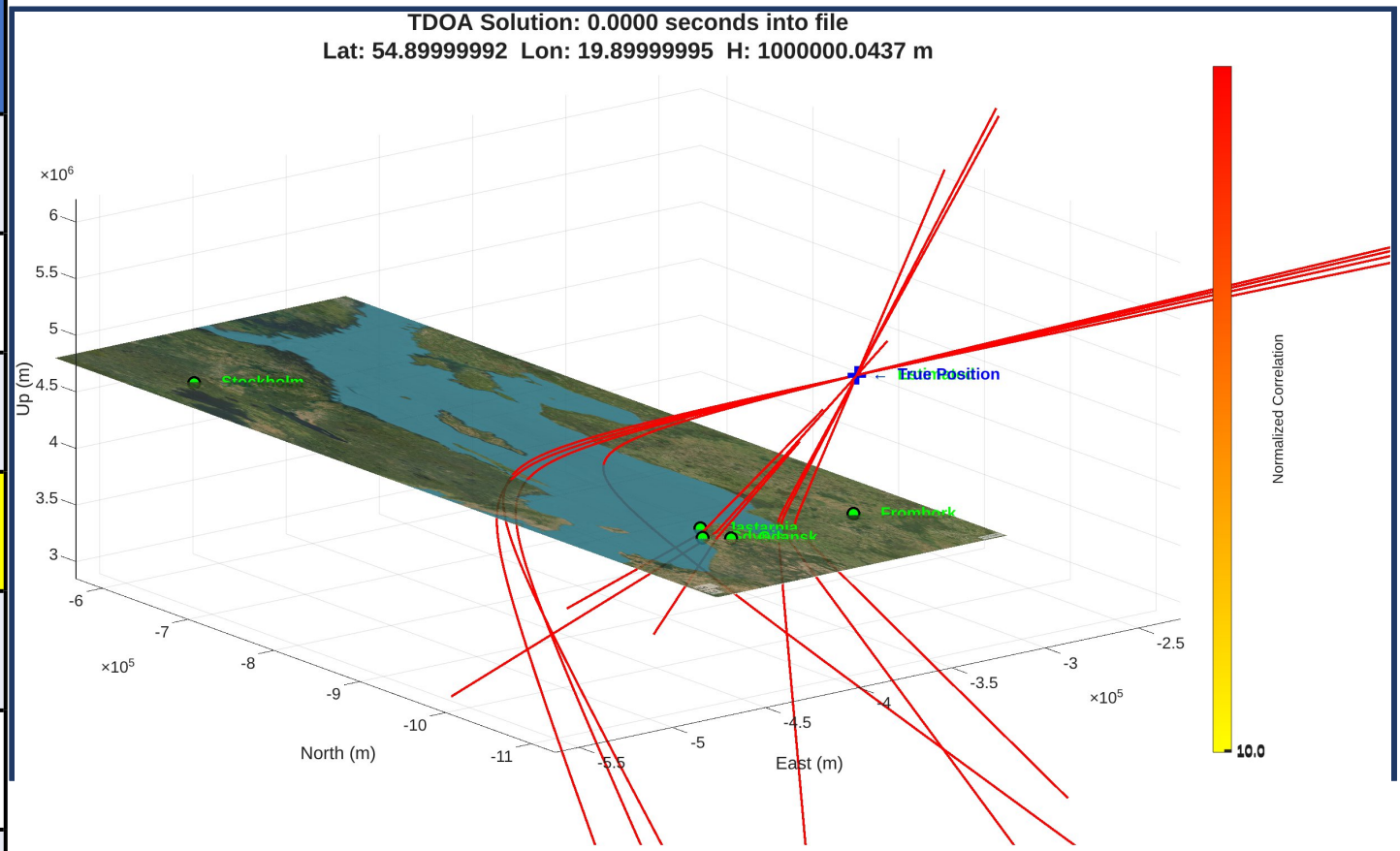


Attempt #1

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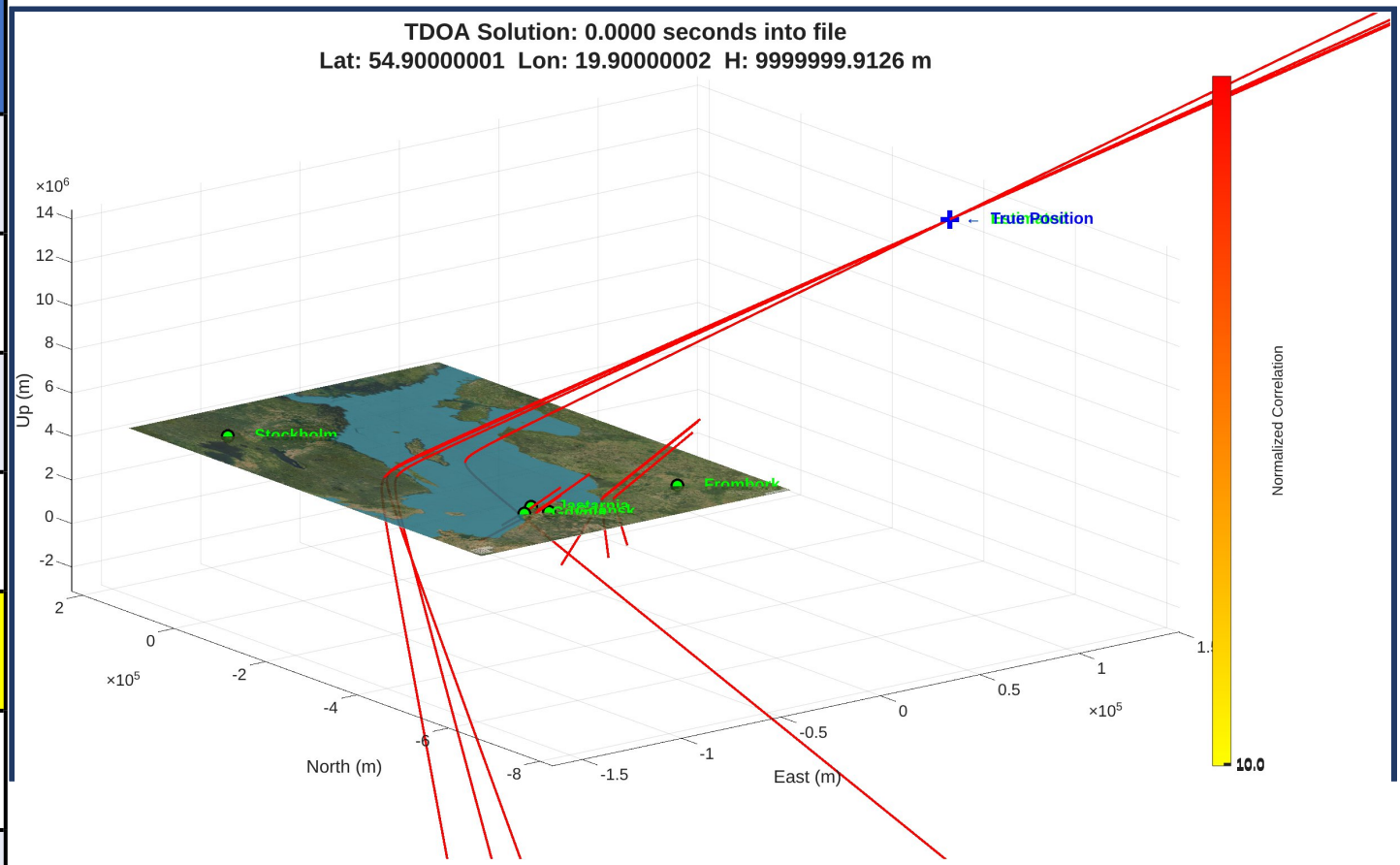


Attempt #1

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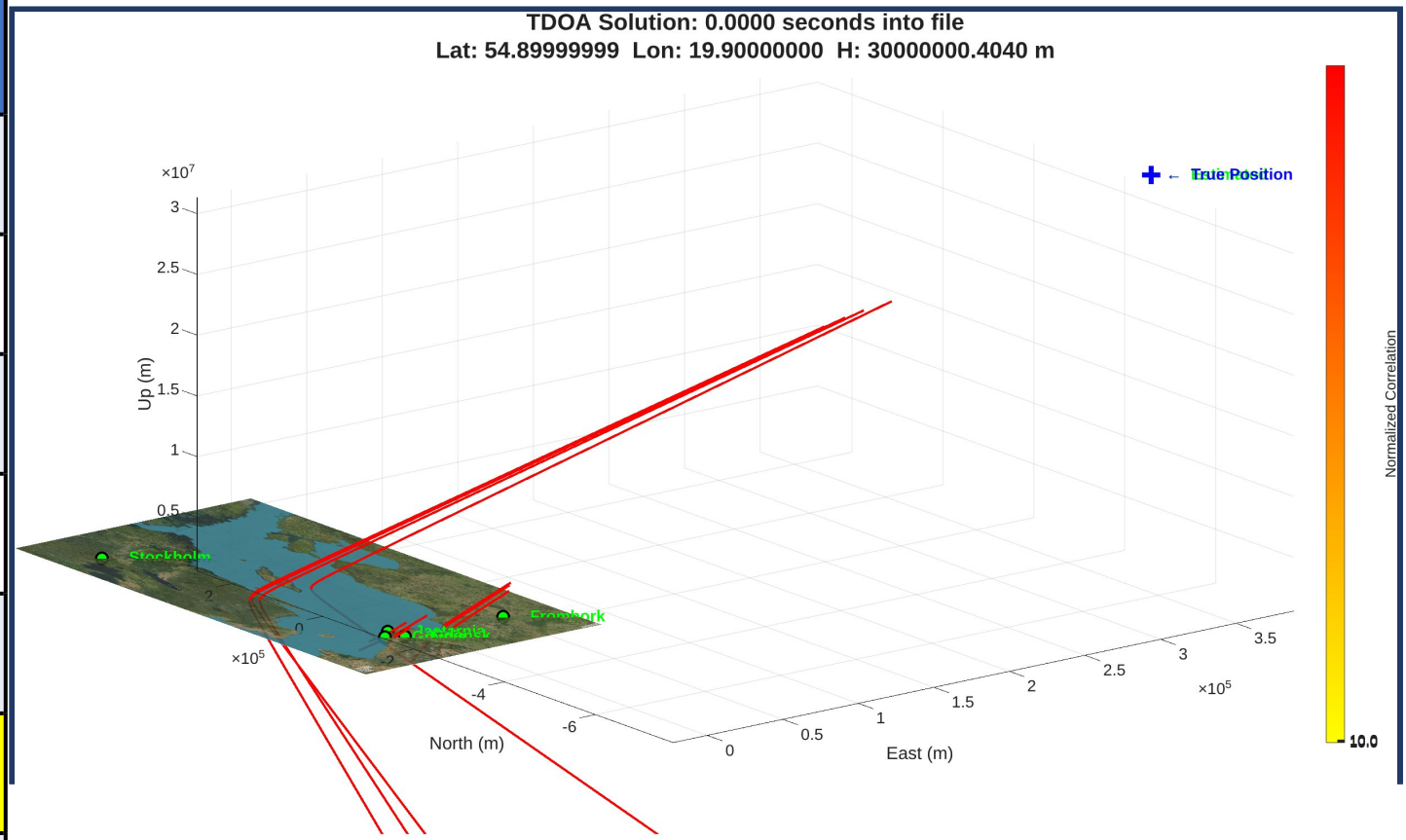


Attempt #1

Results

NLLS with perfect TDOA

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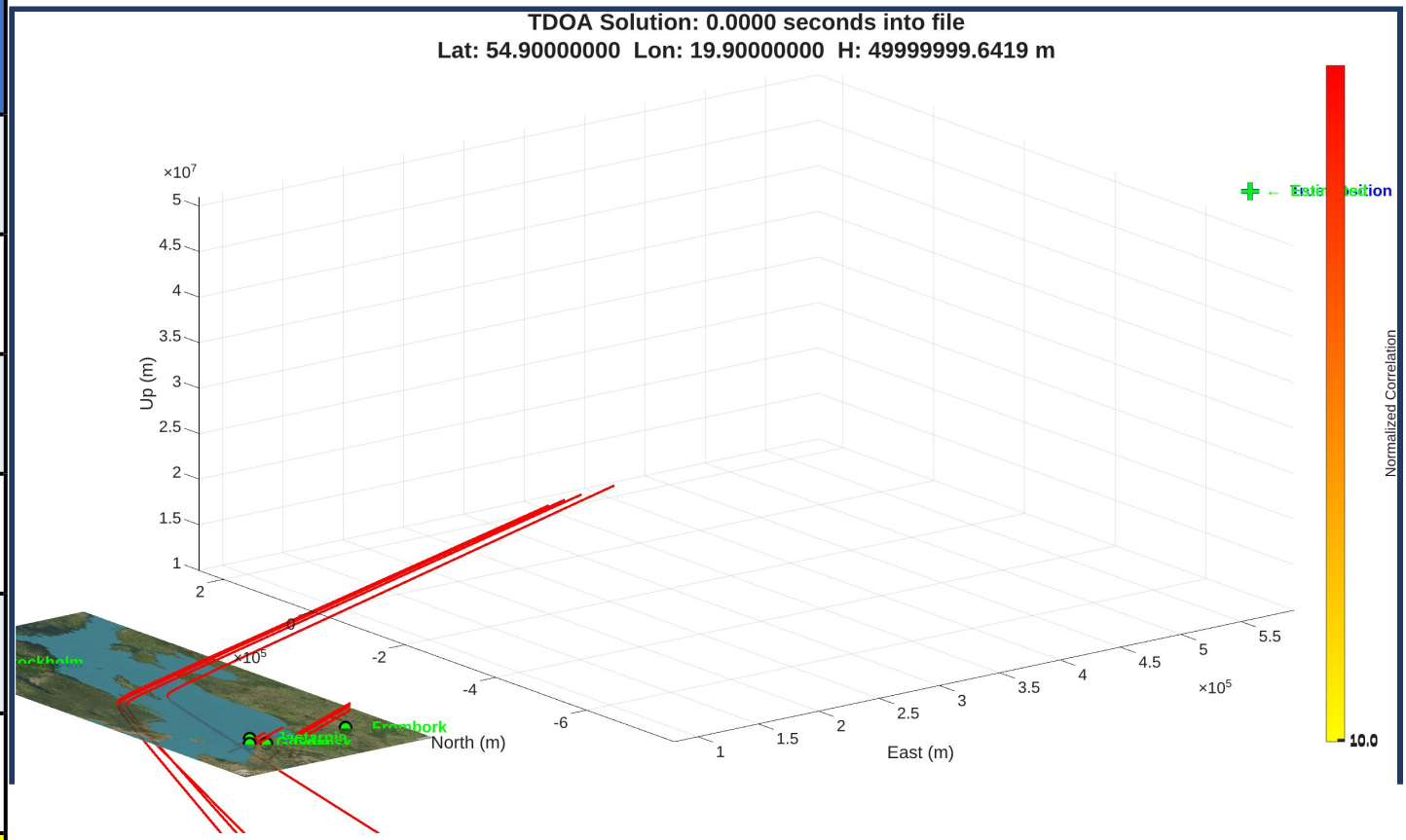


Attempt #1

Results

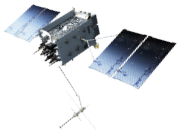
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10,000	10,000	-0.09	0.09
30,000	30,000	0.4	0.4
50,000	50,000	-0.36	0.36



Attempt #1

Method: Part 2: Adding error



Random jitter error:

Local clock of each node is slightly off ➞

Reported signal arrival time is slightly off ➞

TDOA *between* bad clocks is off

To simulate:

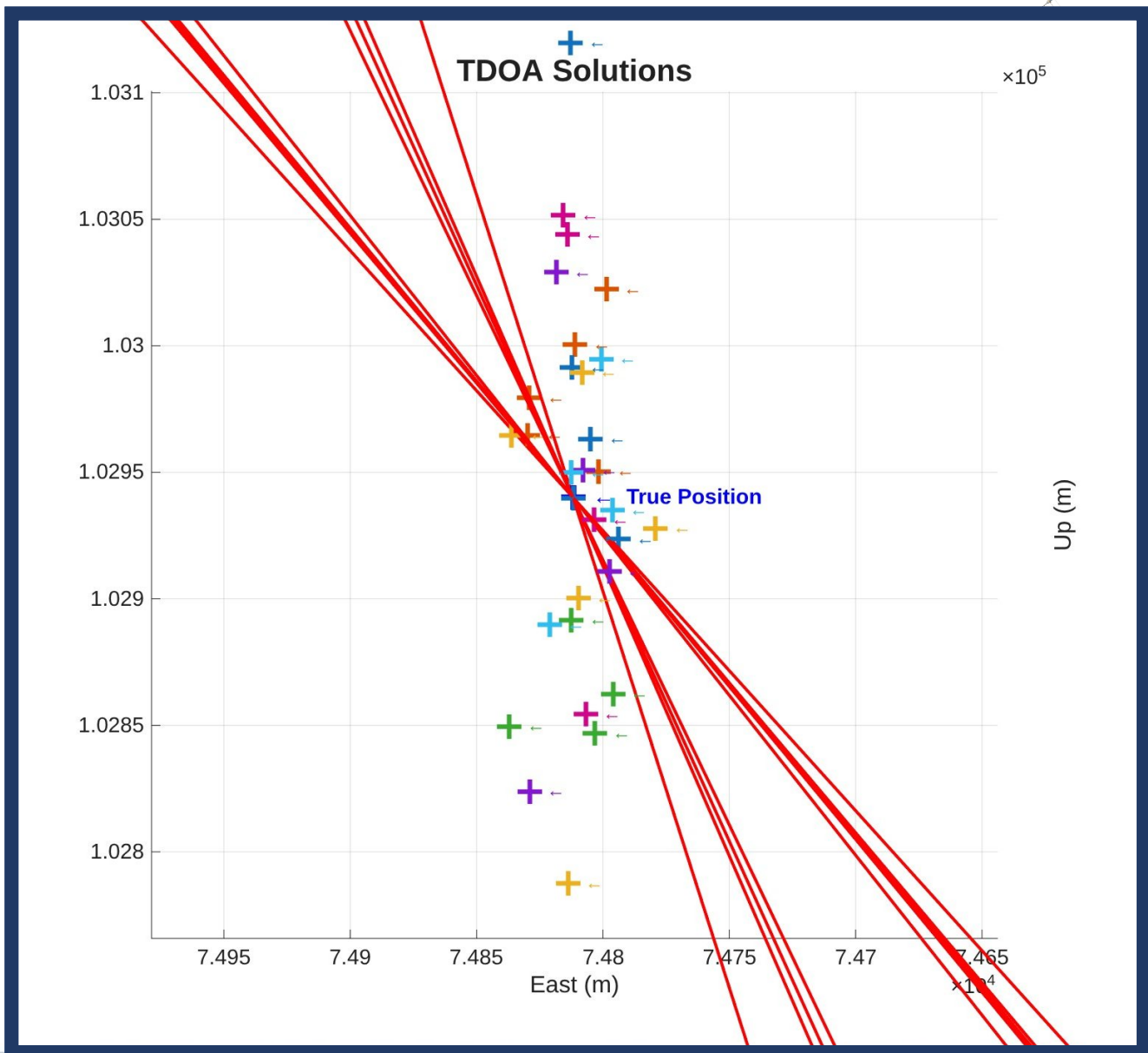
- Find *real* distance of each node to target, convert ➞ time
- Use a standard distribution random number generator
- **~68% of the time, timing error is within +- 20ns** (standard deviation)
- Convert ➞ distance
- TDOA will inherit error from bad distance in each node-pair

Attempt #1

Results

NLLS with 20ns Gaussian timing jitter (30 trials per height)

Height (km)	RMS Horiz. error (km)	RMS Vert. error (km)	V/H error Ratio	Mean Vert. Bias (km)
100	0.03	0.07	2.29	0.00
1000	0.2	0.81	4.04	-0.01
10000	2.07	74.28	35.92	6
30000	6.63	775.53	116.88	106.51
50000	11.2	2133.49	190.57	254.39

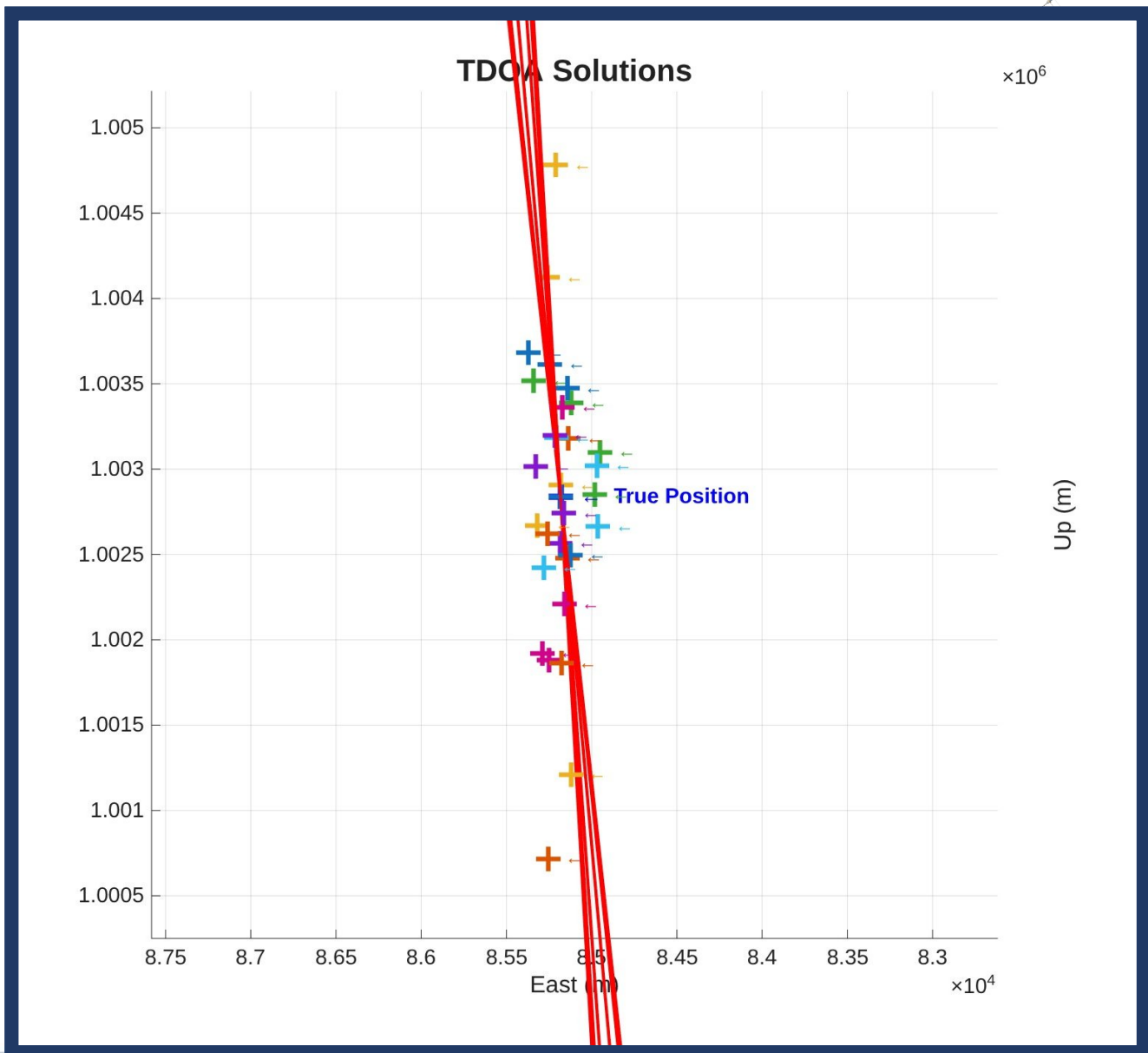


Attempt #1

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30000	6.63	775.53	116.88	106.51
50000	11.2	2133.49	190.57	254.39

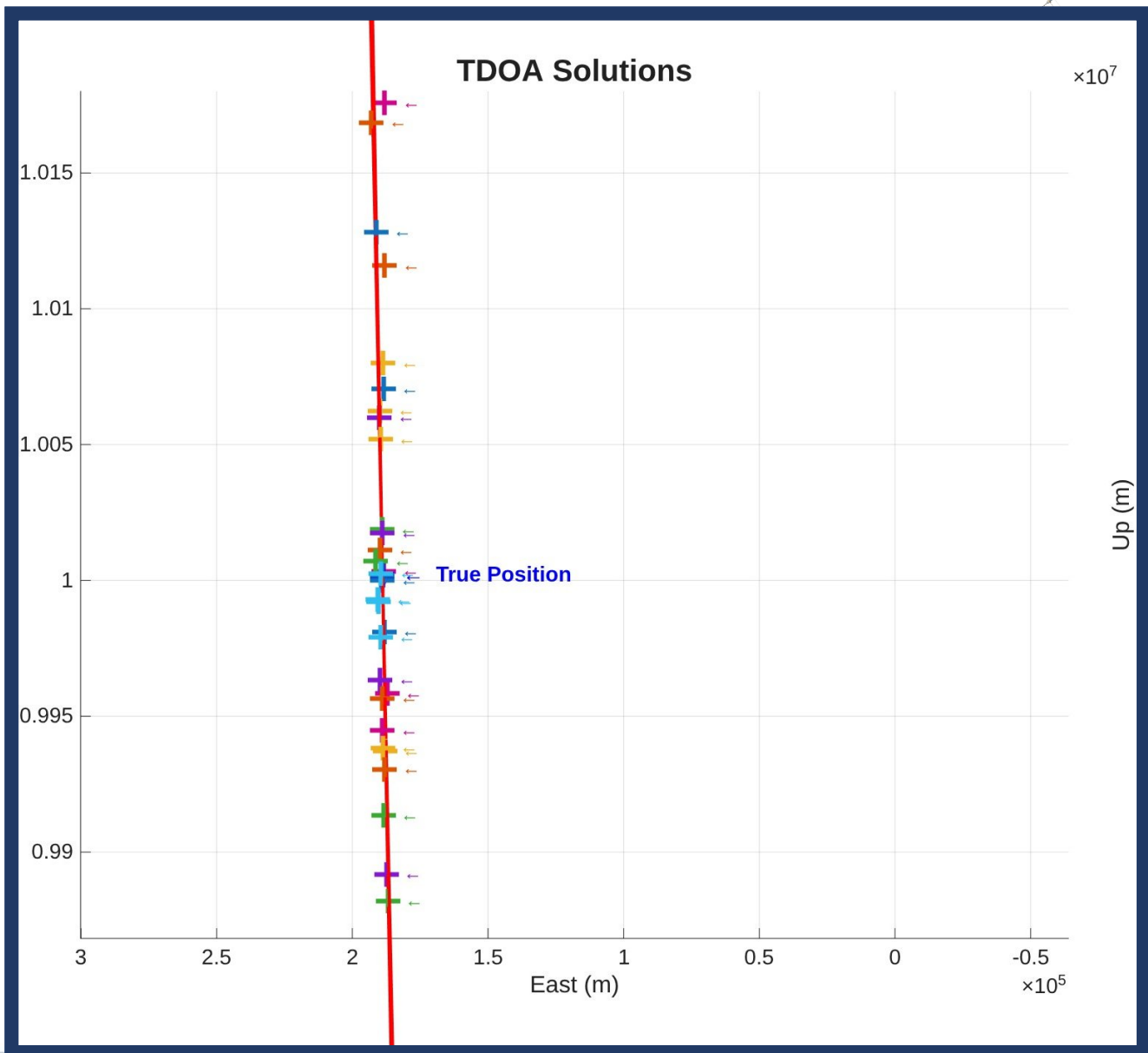


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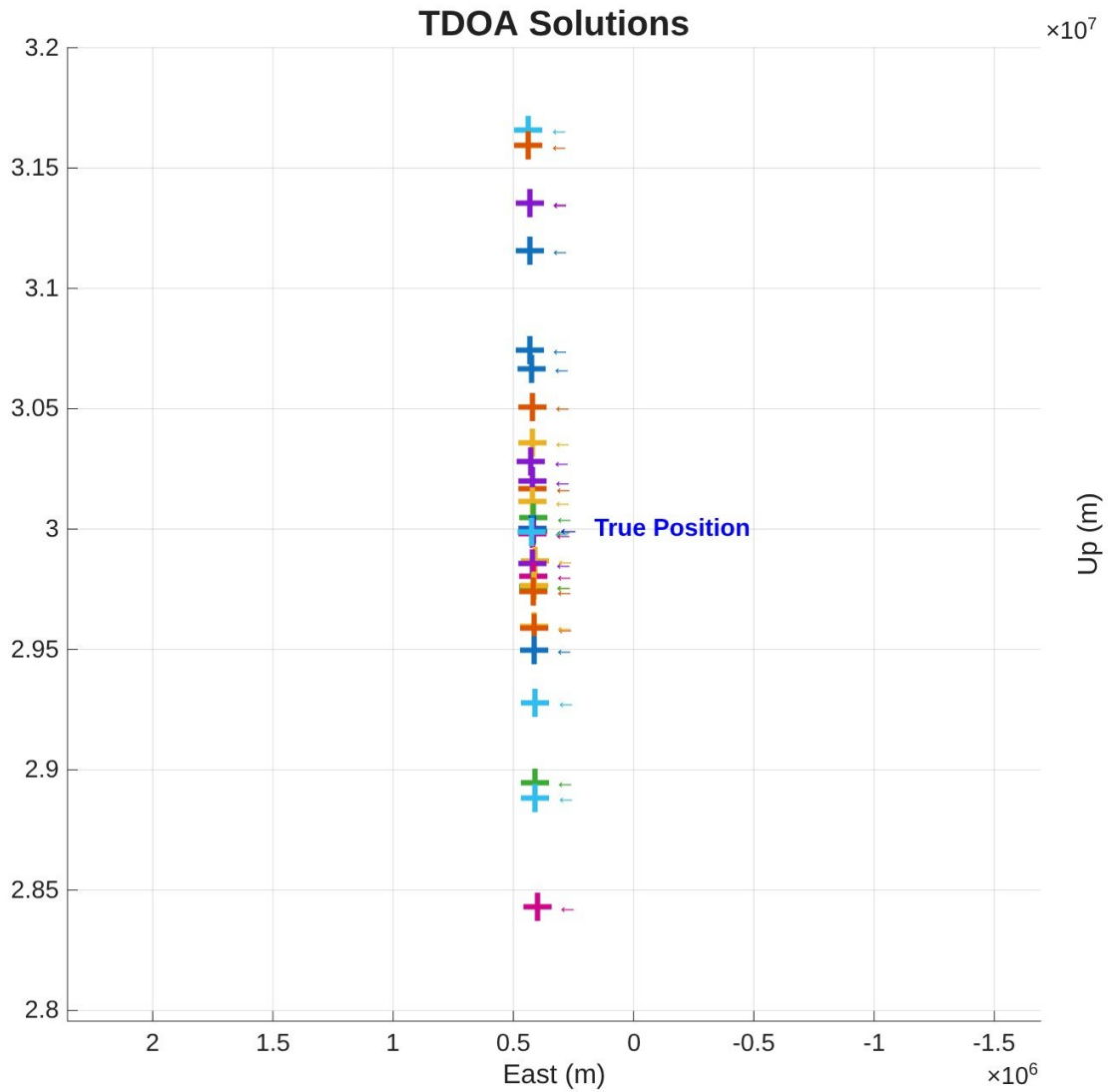


Attempt #1

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Attempt #1

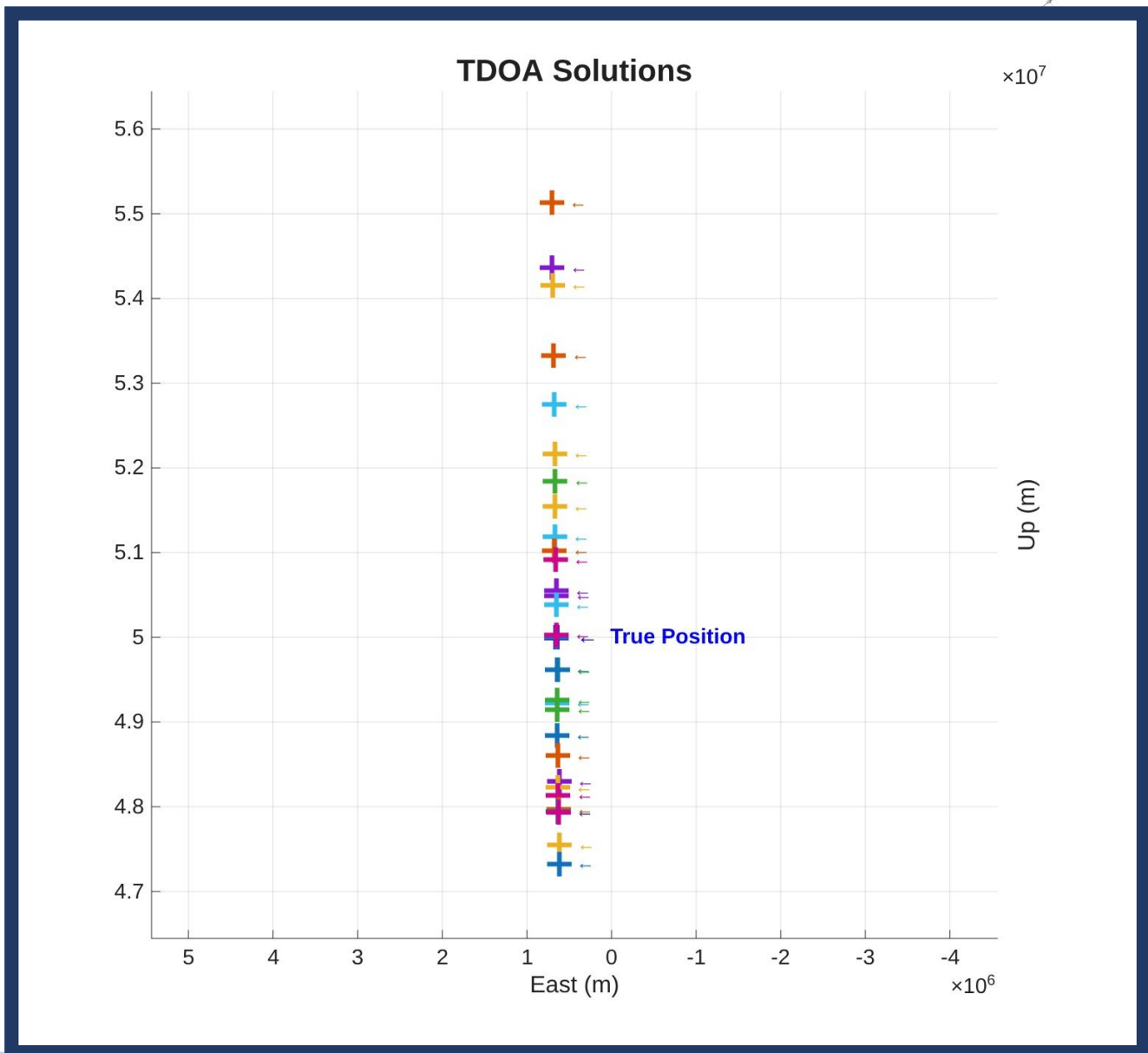
Results

NLLS with 20ns Gaussian timing jitter (30 trials per height)

Notice the VDOP

Height (km)	RMS Horiz. error (km)	RMS Vert. error (km)	V/H error Ratio	Mean Vert. Bias (km)
100	0.03	0.07	2.29	0.00
1000	0.2	0.81	4.04	-0.01
10000	2.07	74.28	35.92	6
30000	6.63	775.53	116.88	106.51
50000	11.2	2133.49	190.57	254.39

(Each height, when given a perfect jitter-free TDOA, yielded 3D error less than 2m)

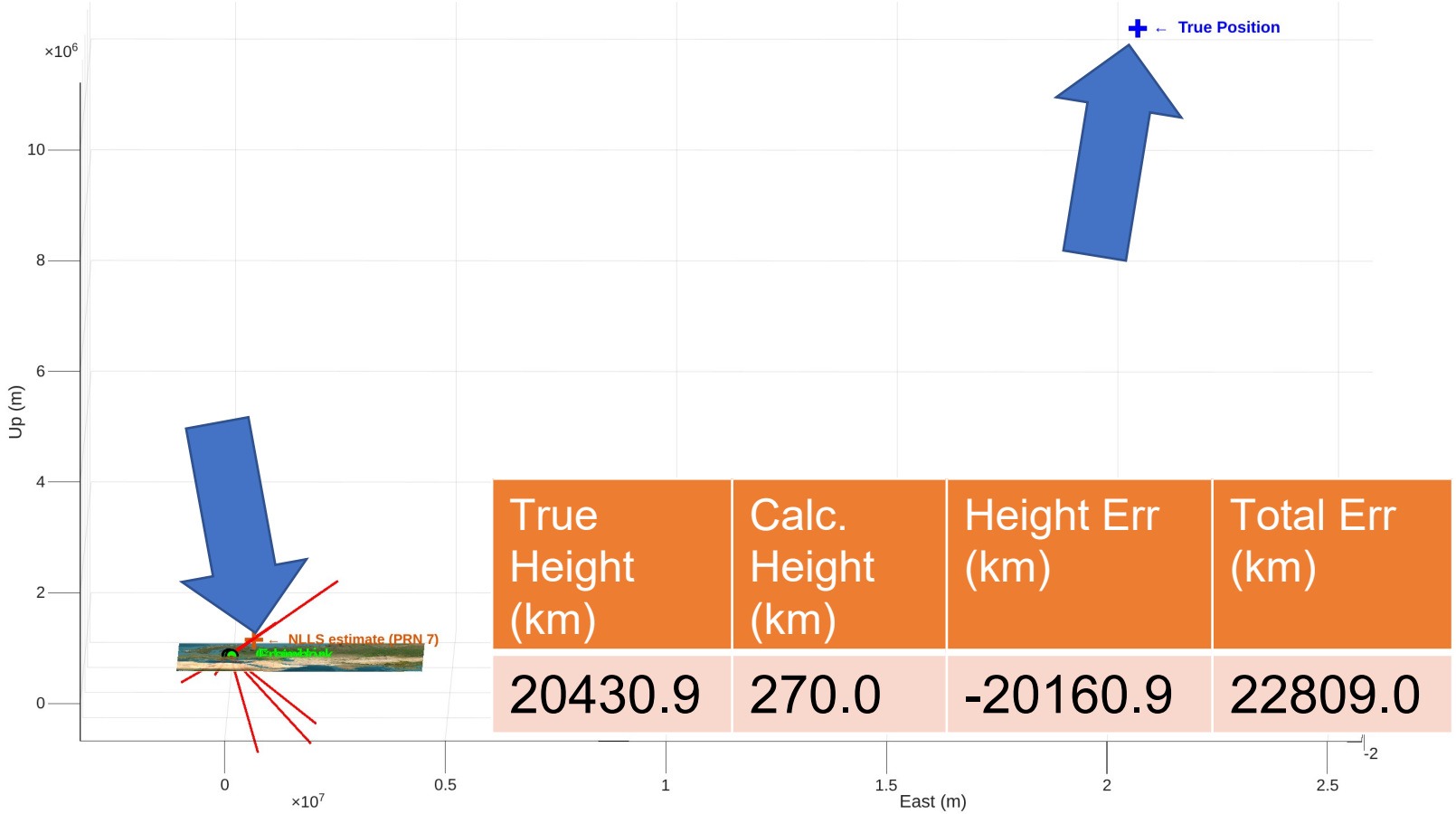


Attempt #1

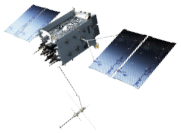
Did our NLLS work on *real* satellites?

No.

- I'm not saying NLLS *couldn't* work.
- NLLS parameters poorly tuned (horizontal search is too small, etc)
- We aren't compensating for atmosphere affecting signal, etc



True Height (km)	Calc. Height (km)	Height Err (km)	Total Err (km)
20430.9	270.0	-20160.9	22809.0

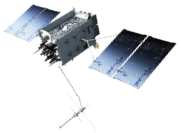


Existing research

How did Clements et al. identify which satellite was causing jamming?

Existing work

A different approach



TDOA + NLLS solving

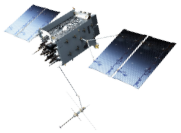
- Solution could be anywhere in 3D space
- Bad geometry ➞ unusable results (bad VDOP)
- Doesn't rely on external data

Catalog-based, ephemeris data TDOA matching

- Reversed logic: Find *theoretical* TDOA of each satellite, compare with observed
- Finite solutions: must match with one satellite in ephemeris data
- “By referencing a catalog of satellite ephemerides .. [they were able to] narrow the possibilities to a manageable few” [2]

Existing work

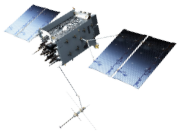
How does this new approach work?



What they did:

- Calculate hypothetical TDOA for each observed GPS satellite
  - Based on final ephemerides (NASA's catalog of satellite orbits)
- Compare to observed TDOAs of PRN signals, identify best matches
  - Based on its regular signal (PRN code)

- Results: "The resulting measured and modeled TDOAs exhibited excellent agreement" [2].

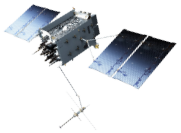


Attempt #2: Catalog matching

Use new approach - ephemeris & PRN codes from **real data** (June 30, Poland, no jamming) to match each signal's TDOA with a catalog.

Attempt #2

Method: TDOA ranking of candidates



4 nodes ➞ 6 node pairs.

Residual aka miss (of each pair) = **Observed** - **Hypothetical** TDOA.

Subtract **mean** of all 6 residuals from each residual (removes any common error)

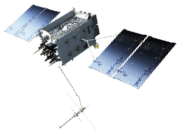
Take **RMS** average of all 6 residuals (so + and – error don't cancel)

Then you get a final **RMS error.**

Higher RMS error ➞ **Worse** agreement between observed and hypothetical TDOA.

Attempt #2
Results

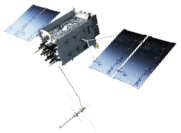
TDOA matching (June 30, Poland)



Test satellite's PRN code	Own RMS error (m)	2nd place PRN	2nd place RMS err (m)	2nd place error÷own error
30	68.5	14	7258.6	106.0x
14	50.5	22	5745.5	113.7x
7	67.1	8	8850.4	131.8x
8	60.8	7	8884.2	146.2x
21	56	5	11000.5	196.4x
22	68.2	14	5816.7	85.3x

We **already know the answer** to which satellite is which because of unique PRN code.

Each satellite's TDOA ranked as its own first place candidate (**success**)!



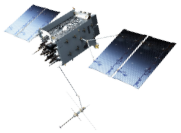
Conclusion

TDOA catalog matching is a better-suited method for localizing satellites versus classic NLLS.

But the paper goes further; in a real jammer situation, they needed to use cross-correlation, and TLEs instead of ephemeris, which would be our next steps.

Conclusion

Future work & implications

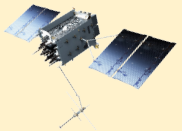


Implications

It is indeed feasible to localize satellites with TDOA, and with future work feasible to localize space jammers. The new paper by Clements et al. illustrates this is a suddenly relevant and pressing concern, into which it is worth devoting more research.

Future work

We'd need a robust cross-correlator regardless of the method we choose. We'll need to use and correct stale TLEs if we want to include all known SVs. We'd need to adapt our ground system to collect rapid C/O drops, or find another source of data.



References

[2] Clements, Zachary L., et al. “Chasing Lightning: Detecting, Characterizing, and Identifying a Powerful Space-Based GNSS Interference Source.” arXiv:2606.03673, arXiv, 2 June 2026. arXiv.org, https://doi.org/10.48550/arXiv.2606.03673.

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Some code (not shown) was generated by Anthropic’s Claude AI LLM, which was also used for brainstorming and planning.